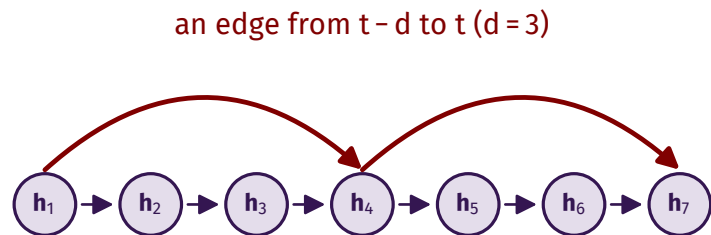
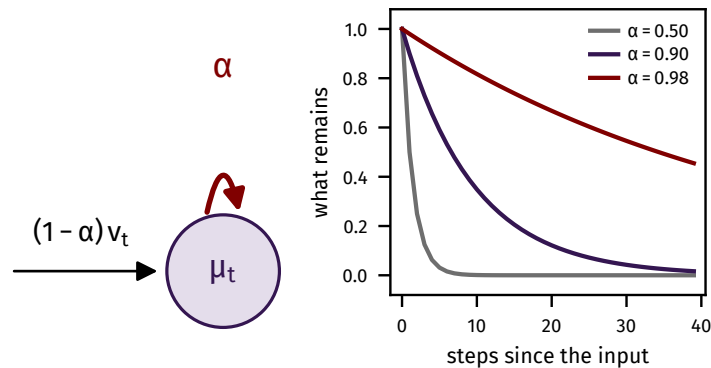


(a) skip connections across time



τ steps along the chain, τ/d along the skips:
the product has τ/d factors instead of τ

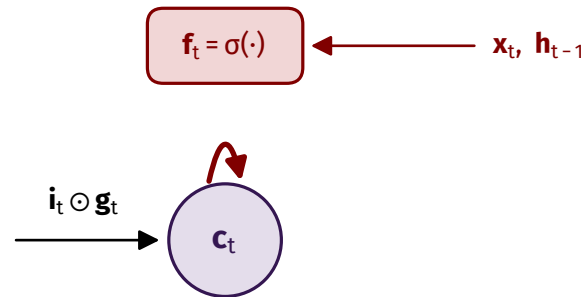
(b) a leaky unit: a fixed self-loop



$$\mu_t = \alpha \mu_{t-1} + (1 - \alpha) v_t$$

$$\partial \mu_t / \partial \mu_{t-1} = \alpha, \text{ a constant near one}$$

(c) a gate: the self-loop's weight is decided by the input



$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$

$$\partial c_t / \partial c_{t-1} = \text{diag}(f_t), \text{ chosen at every step}$$